Hyper-V Lab Report

Task 1: Open Hyper-V Manager and create a new virtual machine named ‘MyTestVM’ with default settings.

Hyper-V Manager was opened and the New Virtual Machine Wizard was used to create MyTestVM. Two values were adjusted from the defaults to suit the laptop:

- Startup memory: set to 2048 MB instead of the 4096 MB shown by default, so the host keeps enough RAM for itself. Dynamic Memory was left on.
- Virtual hard disk: 50 GB, dynamically expanding, stored on drive D:.

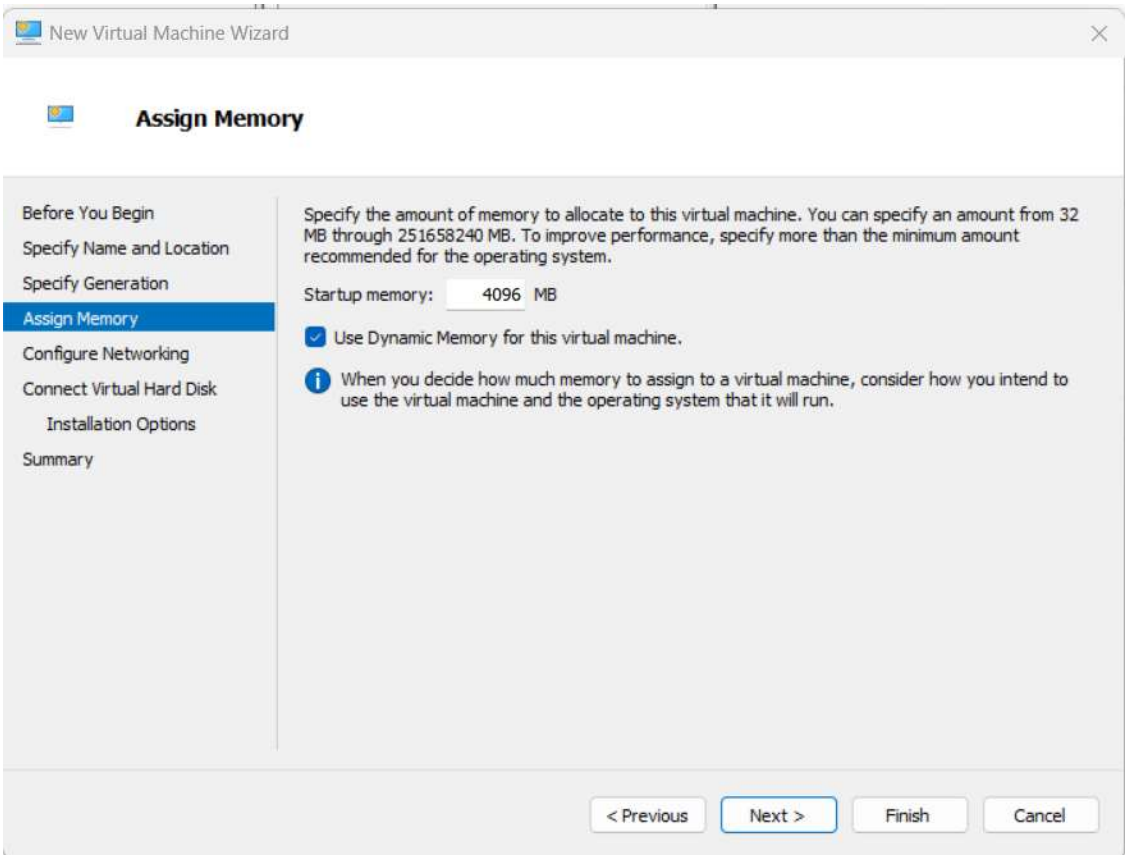


Figure 1: Assign Memory screen of the wizard (default 4096 MB, changed to 2048 MB)

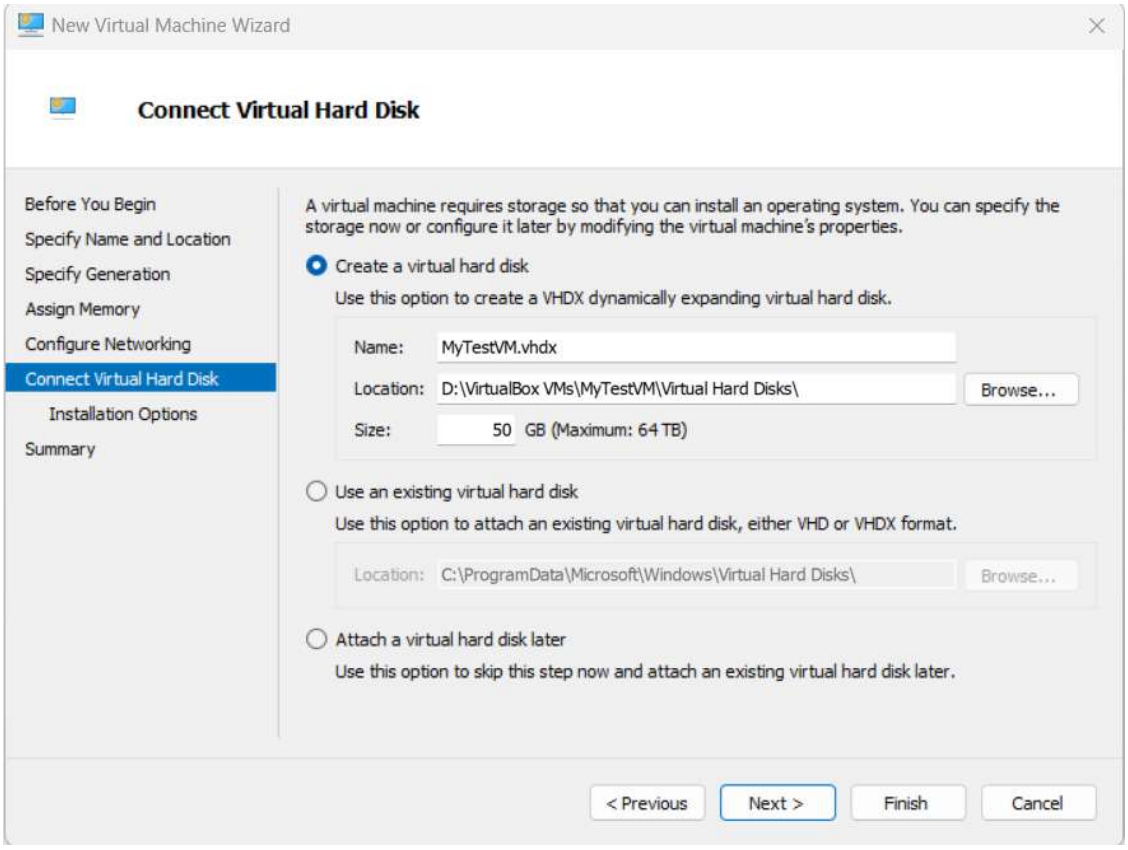


Figure 2: Connect Virtual Hard Disk screen (MyTestVM.vhdx, 50 GB)

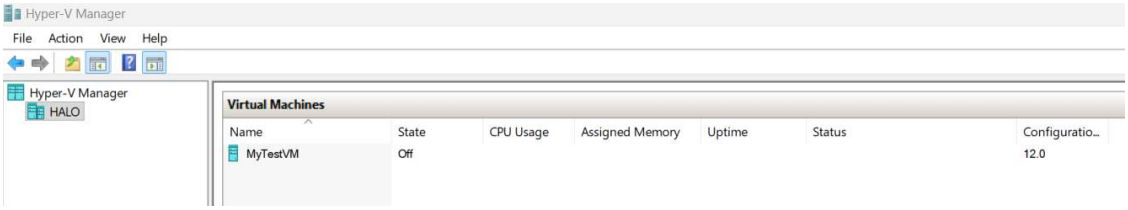


Figure 3: MyTestVM listed in Hyper-V Manager on host HALO

Task 2: Configure a virtual switch that allows your virtual machines to access the internet, and attach 'MyTestVM' to this switch. (Hint: use the Virtual Switch Manager and select 'External'.)

In Virtual Switch Manager a new switch named ExternalSwitch was created with connection type External, bound to the Realtek RTL8821CE 802.11ac Wi-Fi adapter. The option "Allow management operating system to share this network adapter" was ticked so the host keeps its own internet connection. MyTestVM was then shut down, its Network Adapter was set to ExternalSwitch under Settings, and the VM was started again.

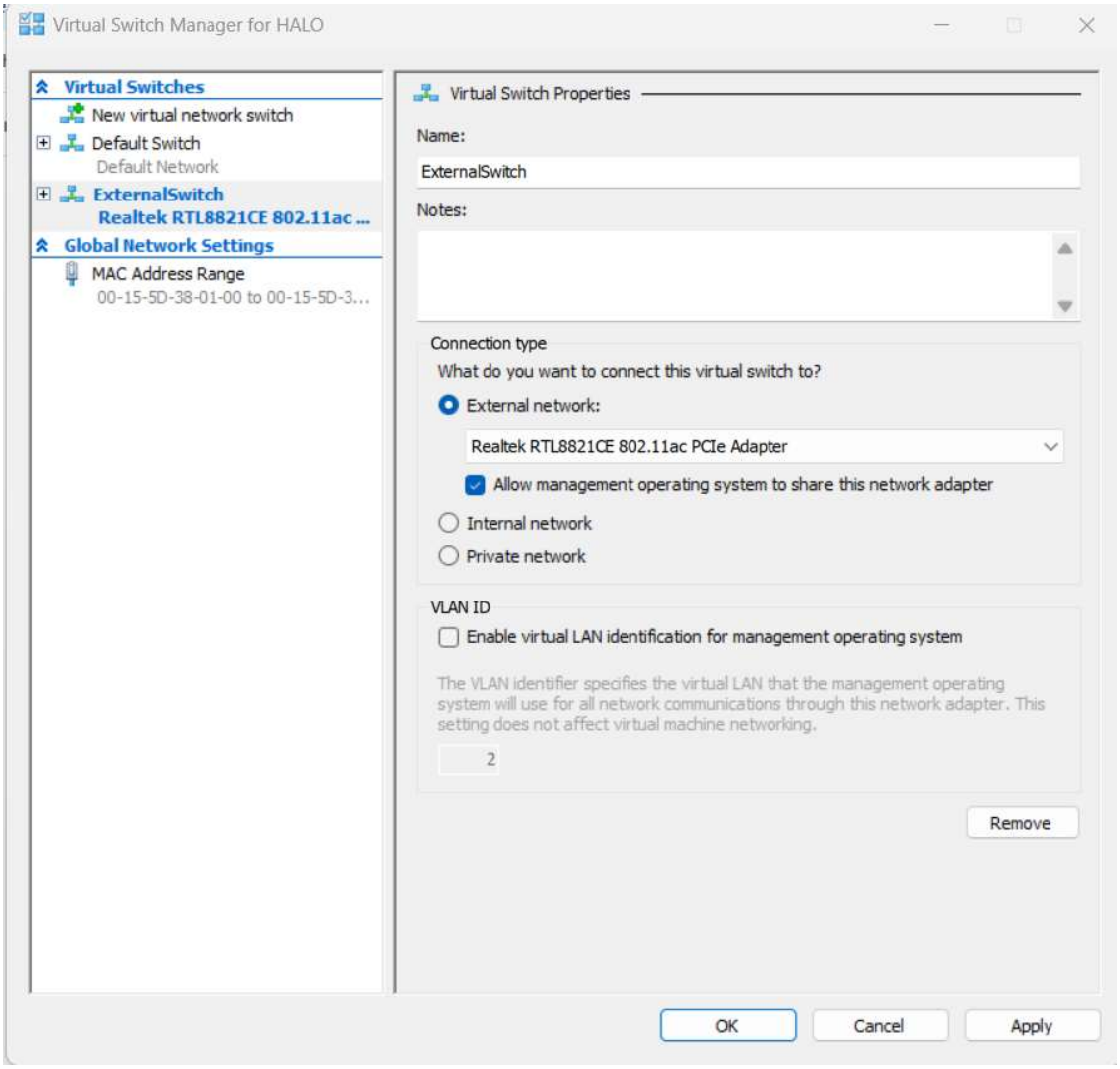


Figure 4: ExternalSwitch settings (External network, Realtek Wi-Fi adapter)

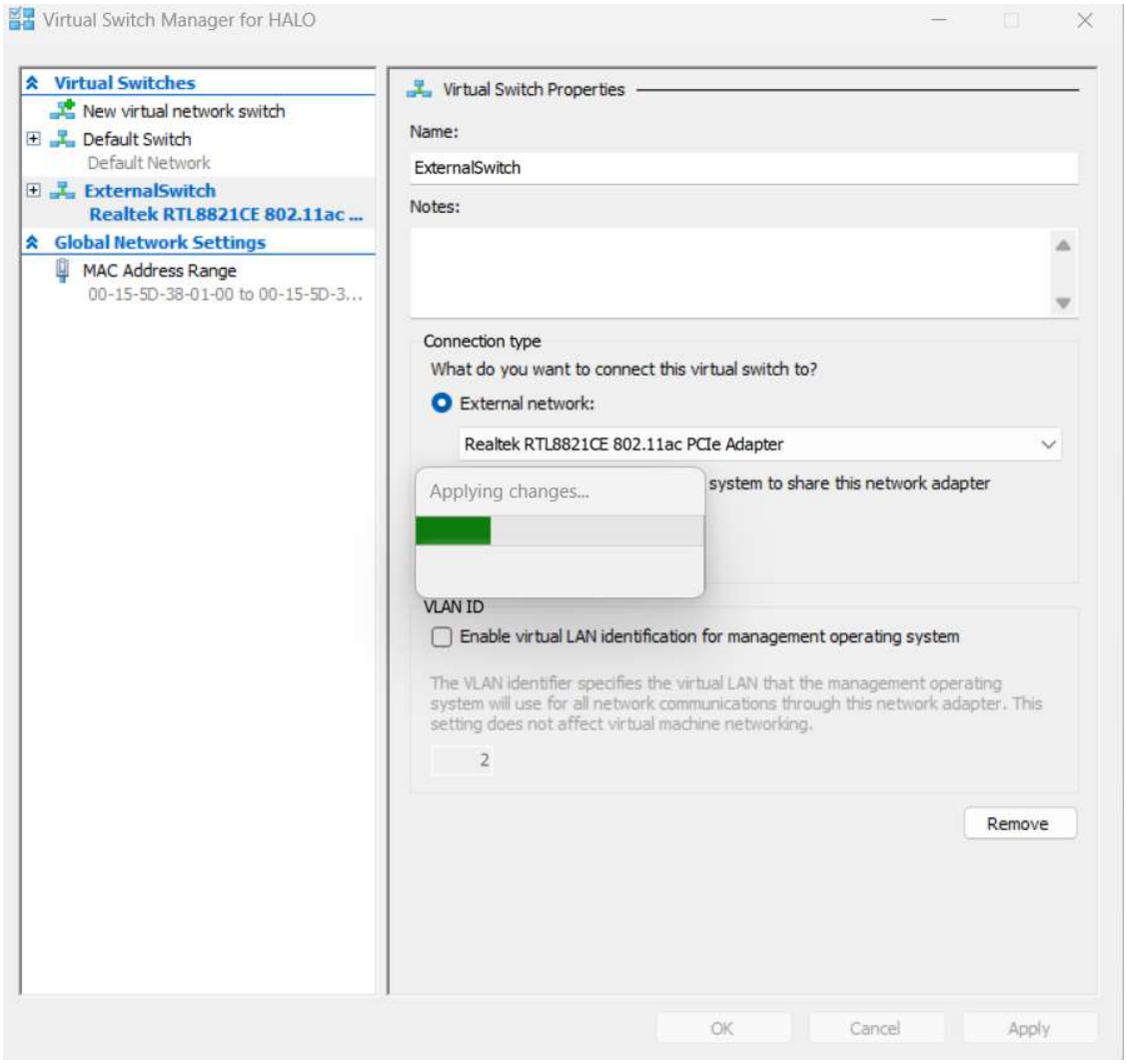


Figure 5: Hyper-V applying the virtual switch changes

To confirm internet access, Microsoft Edge was opened inside MyTestVM and wikipedia.org loaded successfully.

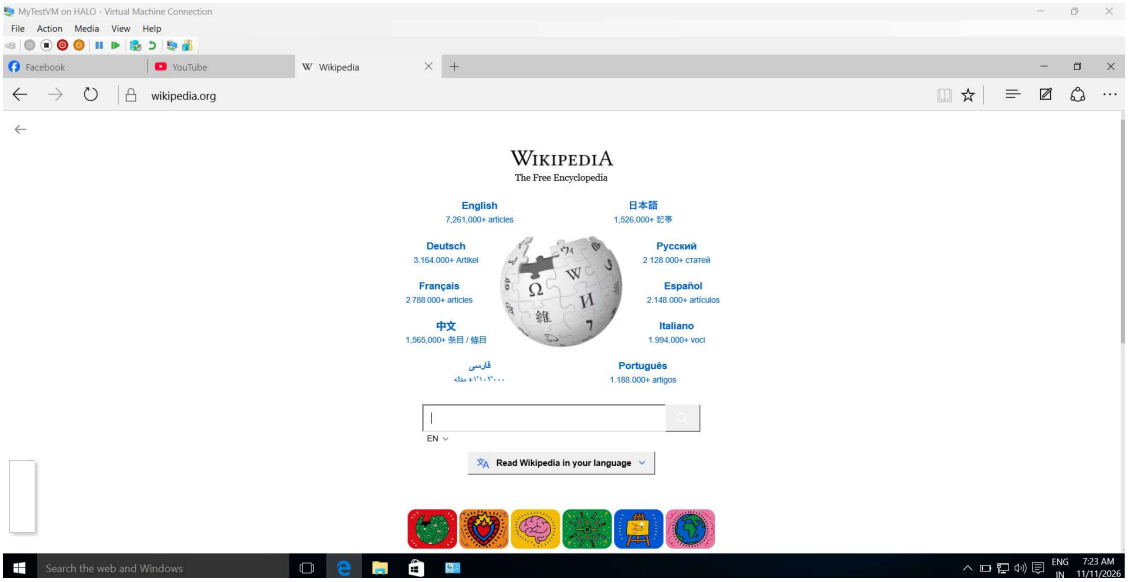


Figure 6: MyTestVM browsing wikipedia.org through ExternalSwitch

Task 3: Install a Windows 10 ISO on ‘MyTestVM’ and ensure it boots successfully to the desktop.

The Windows 10 ISO was attached to the VM and the VM was started. Windows Setup ran, the installation completed, and the VM restarted from its virtual disk. The final setup step created a local account, after which the VM reached the Windows 10 desktop.

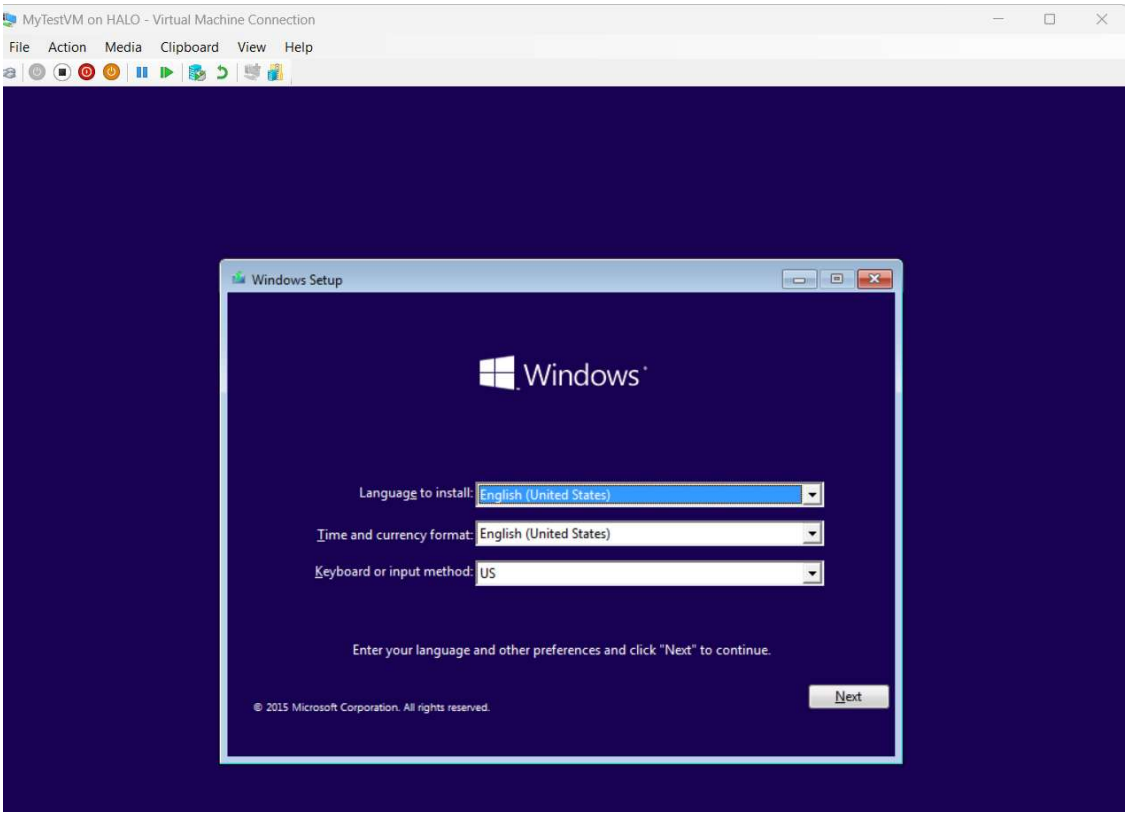


Figure 7: Windows Setup booting from the ISO

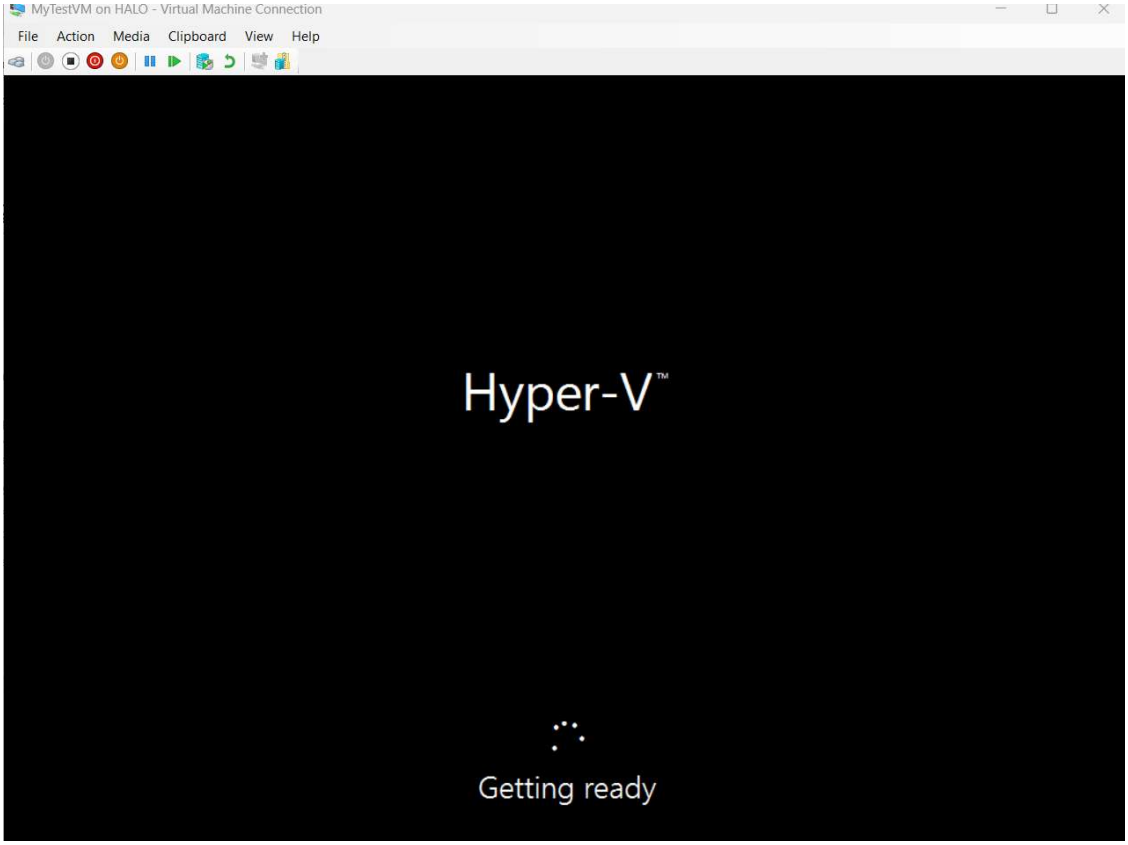


Figure 8: First boot from the virtual disk ("Getting ready")

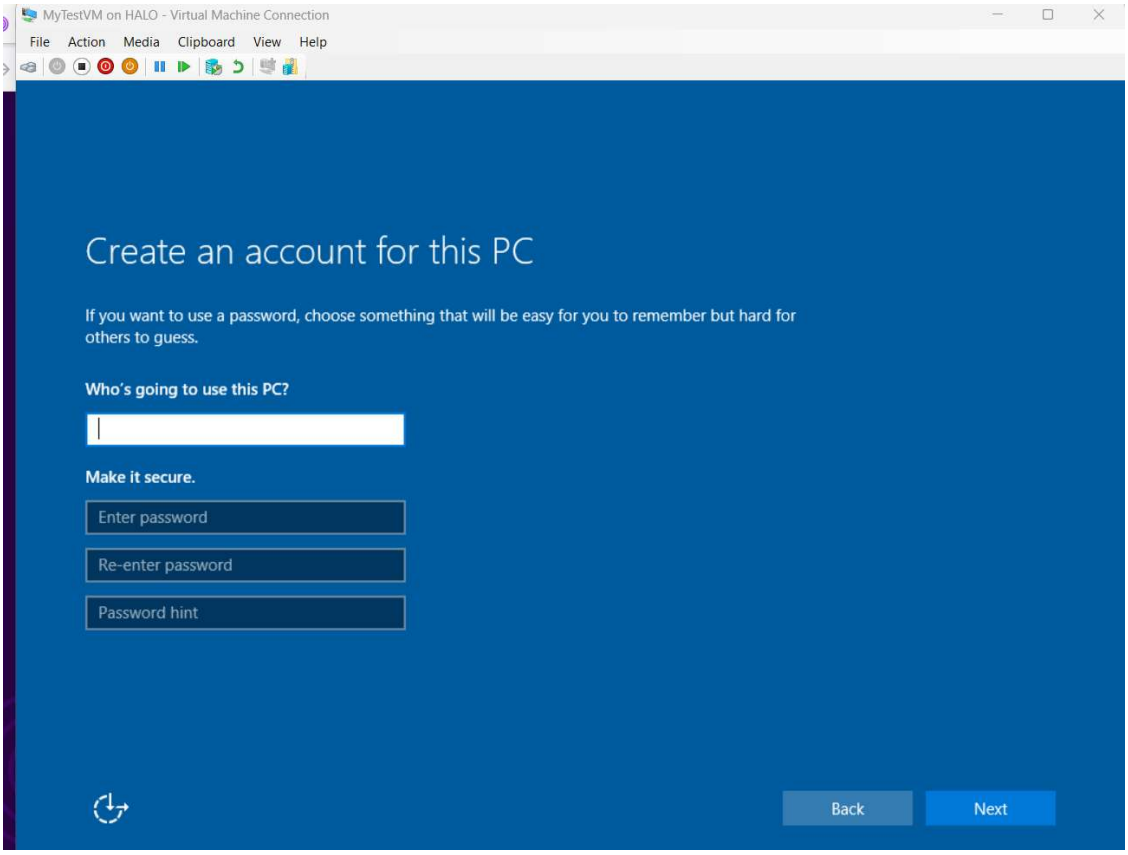


Figure 9: Creating a local account at the end of setup

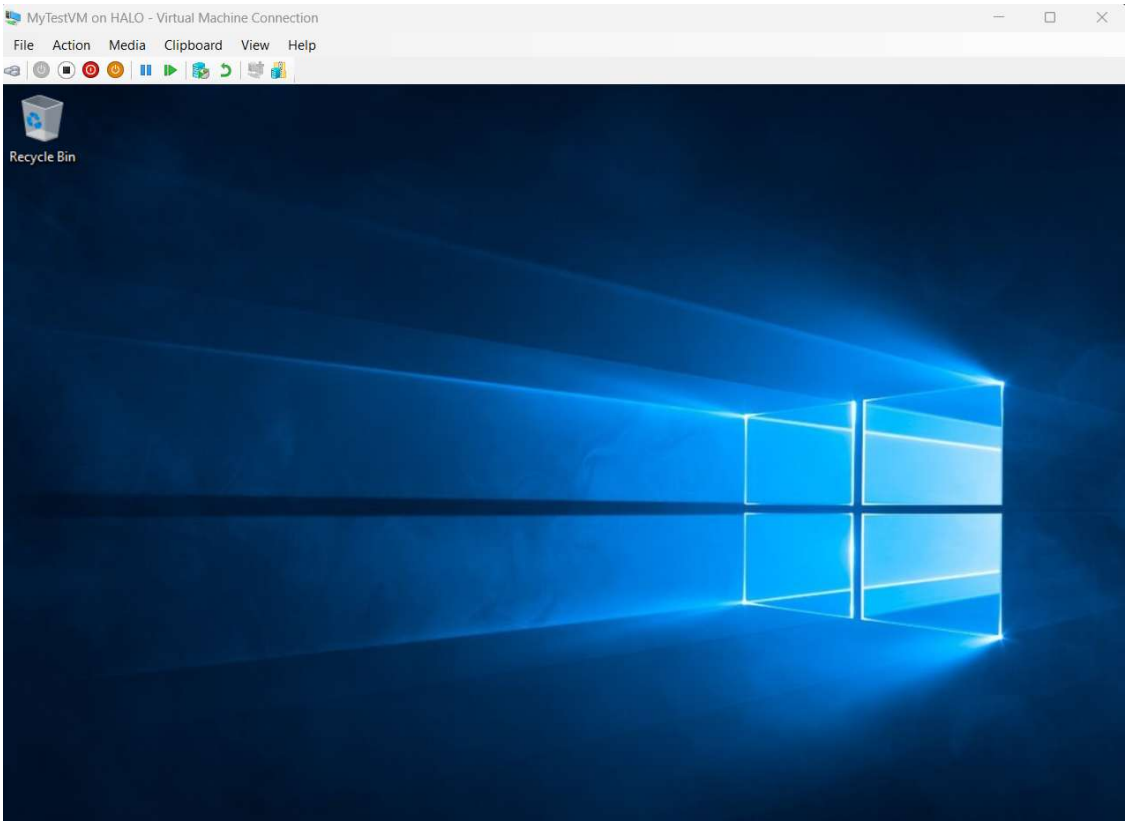


Figure 10: Windows 10 desktop in MyTestVM on HALO

Task 4: Take a snapshot (checkpoint) of 'MyTestVM' after installation, then make a change inside the VM (like creating a folder on the desktop), and restore the VM to the checkpoint to verify the change is undone.

Steps performed:

- With the VM on the desktop, Action > Checkpoint was used to create a checkpoint named AfterInstall.
- A new folder was created on the VM desktop.
- Action > Revert was chosen and the revert prompt was confirmed.
- The VM restarted from the saved state and the new folder was gone.

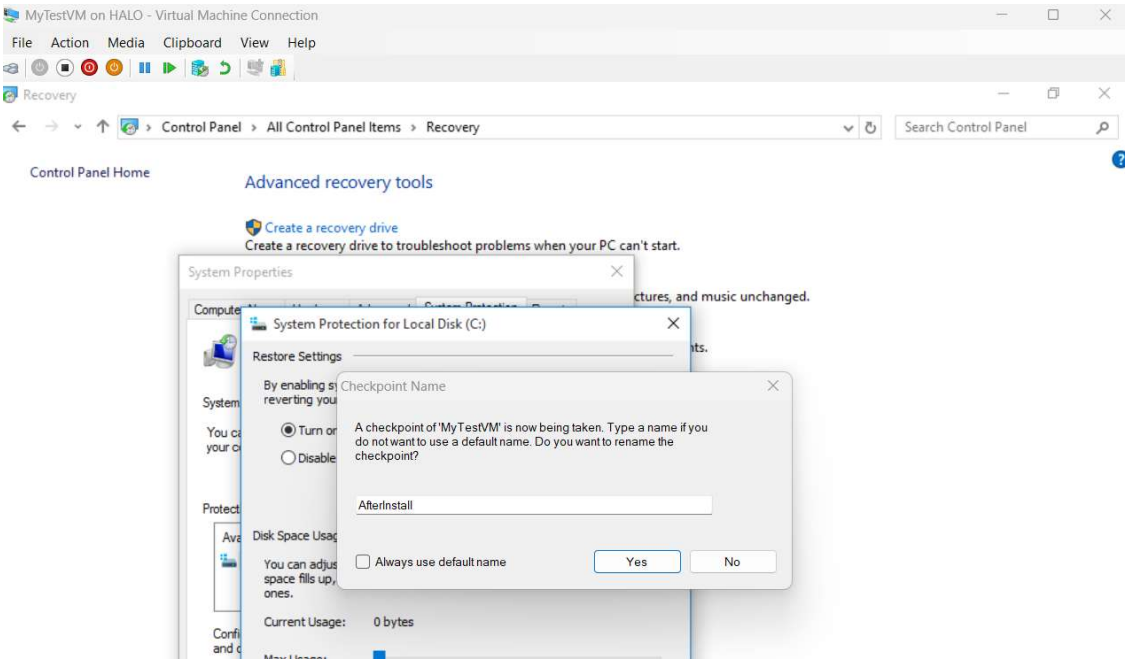


Figure 11: Creating the checkpoint named AfterInstall

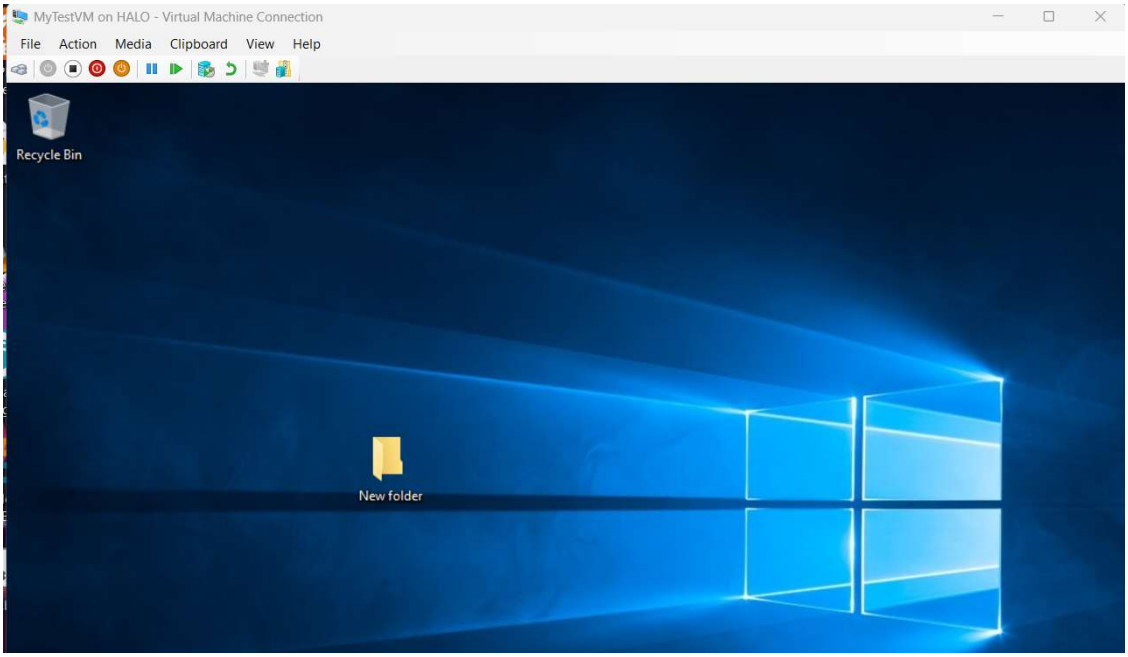


Figure 12: "New folder" created on the desktop after the checkpoint

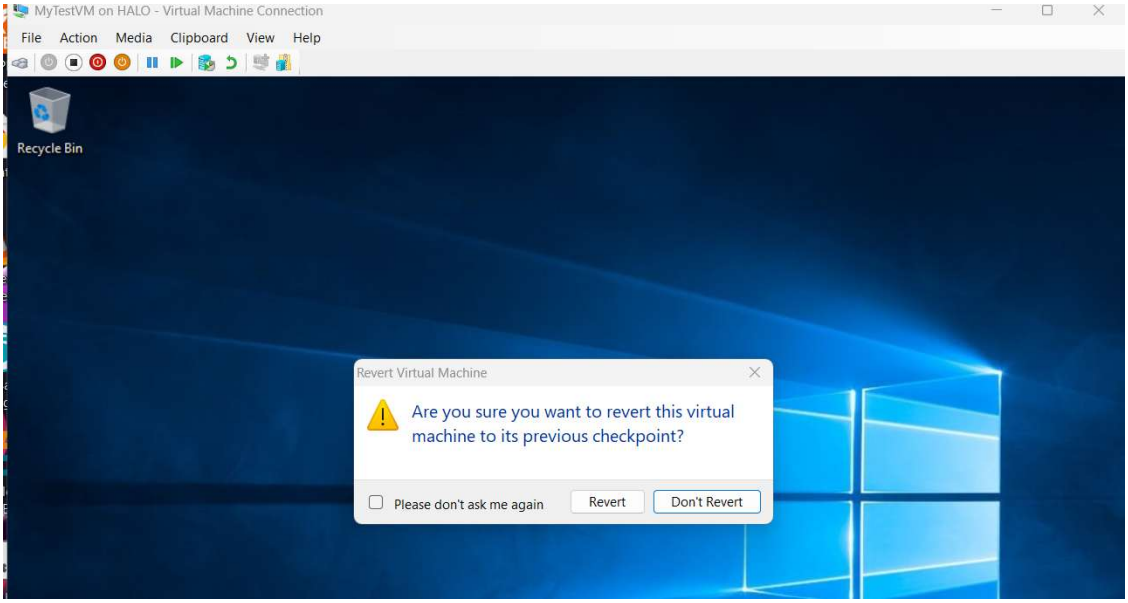


Figure 13: Revert confirmation prompt

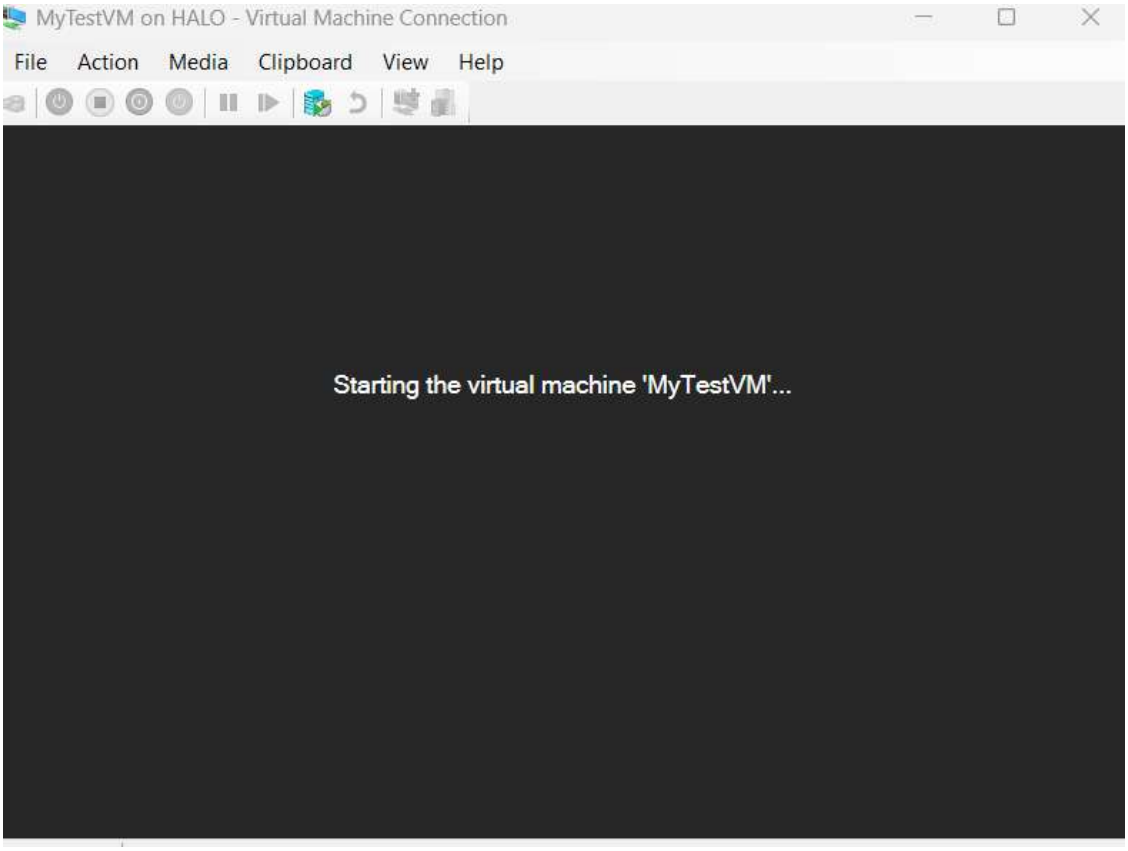


Figure 14: VM restarting from the checkpoint

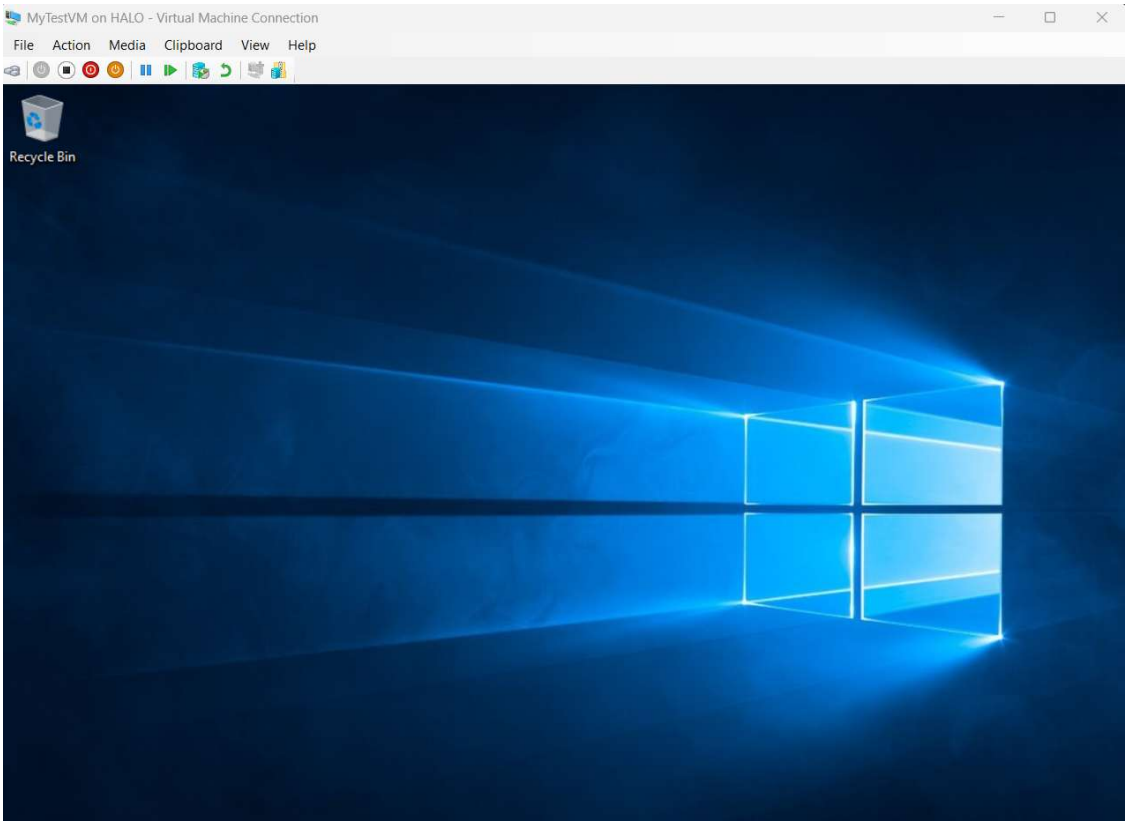


Figure 15: Desktop after the revert, with the folder removed

Task 5: Use ChatGPT or Copilot to generate PowerShell commands for creating a new Hyper-V VM with 2GB RAM and a 40GB disk, then run the commands and verify the VM appears in Hyper-V Manager.

An AI assistant was asked for PowerShell commands to create a Hyper-V VM with 2 GB RAM and a 40 GB disk. The commands below were run in PowerShell (Administrator) on the Windows 11 host. The folder is created first, then the VM, and Get-VM lists it.

```
New-Item -ItemType Directory -Path "D:\Hyper-V" -Force

New-VM -Name "PSTestVM" -MemoryStartupBytes 2GB -Generation 2 `
  -NewVHDPath "D:\Hyper-V\PSTestVM.vhdx" -NewVHDSizeBytes 40GB `
  -SwitchName "ExternalSwitch"

Get-VM -Name "PSTestVM"
```

The same commands were then run in PowerShell (Administrator) on the Windows 11 host. The D:\Hyper-V folder was created and Get-VM shows PSTestVM, Generation 2, State Off, Status “Operating normally”.

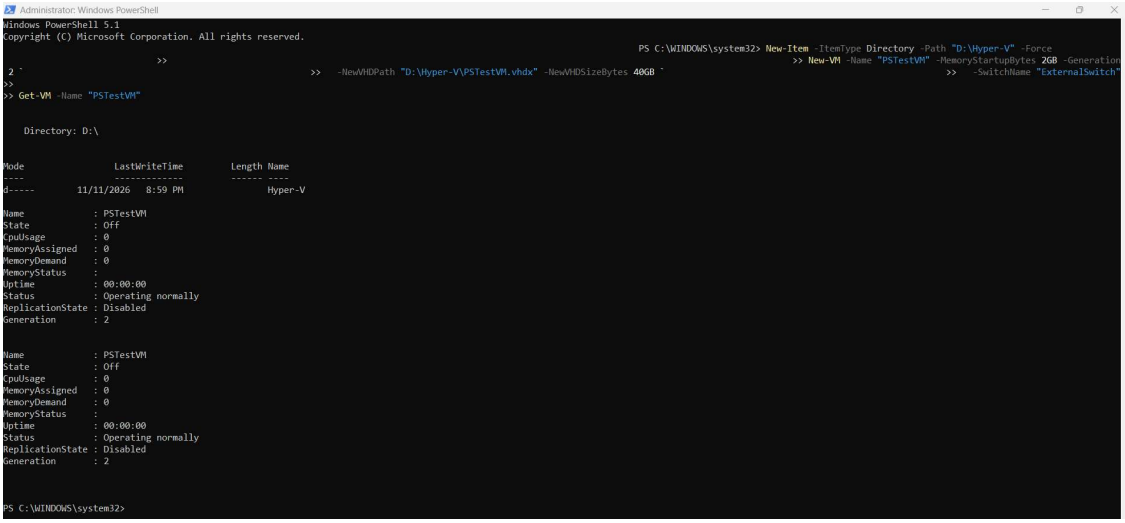


Figure 16: PowerShell output on the host showing PSTestVM

Hyper-V Manager lists PSTestVM next to MyTestVM. Its settings show 2048 MB RAM, the PSTestVM.vhdx hard drive and ExternalSwitch as the network adapter.

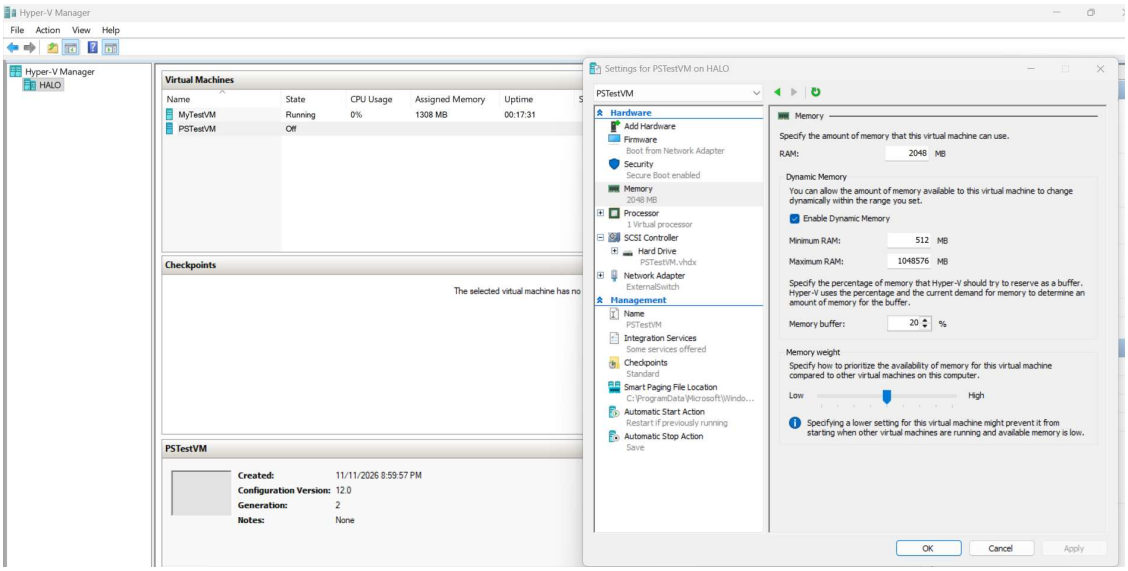


Figure 17: PSTestVM in Hyper-V Manager with its memory settings